

Ilia Malkin

Junior Python Engineer · LLM integrations & backend automation

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SUMMARY

Backend-focused Python engineer with a Data Science background (Universität Wien), building production LLM pipelines, agentic systems and multi-provider LLM routers with fallback and circuit breaker logic. No team experience yet; five open-source projects of varying depth back the resume instead. Strongest in LLM SDK integration (OpenAI, Gemini, xAI/Grok, MiMo, OpenRouter), async Python, prompt engineering, Pydantic-typed structured outputs and multi-provider fallback patterns. Looking for junior backend / AI-engineering roles, remote or Vienna-based.

PROJECTS

Stager · multi-tenant expense tracker

May 2026 · github.com/kudnever/stager

Telegram bot + FastAPI backend + Next.js admin for receipt-photo OCR via LLM

- Designed and shipped a 9-service Docker Compose stack (bot, api, worker, beat, postgres, redis, minio, web, caddy) on a single Hetzner VPS; full Alembic migrations, healthchecks, Caddy TLS.
- Built a custom LLM router (MiMo → Gemini fallback) with Redis-based soft circuit breaker, Pydantic-validated structured output, structlog metrics and per-request fallback-reason logging.
- Backend in FastAPI + Pydantic v2 + async SQLAlchemy 2.0 (PostgreSQL 16); JWT auth + RBAC (owner/editor/viewer); invite-token redeem flow.
- Bot in aiogram 3 with FSM stored in Redis: photo → OCR → review-card → inline-button edits → expense persistence.
- Async jobs via Celery + Redis (OCR, XLSX export with openpyxl); Celery Beat for scheduled tasks.
- Web admin in Next.js 14 (App Router) + Tailwind + recharts, JWT in httpOnly cookie, server actions for mutations.
- Test stack: pytest + testcontainers (real Postgres + Redis in CI) + respx for LLM mocking; GitHub Actions runs ruff + mypy strict + tests on every PR; pre-commit + gitleaks locally.

doc-assistant-bot · production RAG service

2025 · github.com/kudnever/doc-assistant-bot

Telegram bot that answers questions over uploaded documentation

- RAG pipeline on FastAPI + SQLite with sqlite-vec for vector search and fastembed for embeddings; OpenRouter multi-provider LLM routing.
- Background ingestion worker with pytest-tested guards against malformed inputs; Dockerised deployment.

job-hunt eval framework

May 2026 · research / automation

Eval framework and production pipeline for auto-generating cover letters against hh.ru job postings

- Pipeline: pull → dedup → score → filter → employer-research → generate; structured retry guard with one-shot regeneration on format violation; per-vacancy variant selection.
- A/B tested gpt-5.5-codex against MiMo V2.5-Pro across reasoning_effort tiers; measured cached_tokens, reasoning_tokens, latency per call.
- Achieved 5× cost reduction (~\$0.009 per letter, 4 to 7 seconds latency) while passing all hard-rule format checks (banned openers, AI-tells, length bounds, quote constraints).

revenue-metrics-console

2025 · github.com/kudnever/revenue-metrics-console

Revenue dashboard frontend with custom filters and chart components

- React 18 + TypeScript + Tailwind + TanStack Query for server state; custom period filters, recharts visualisations, loading/error/empty state handling.

windows-claude-code-doctor

2026 · github.com/kudnever/windows-claude-code-doctor

Claude Code skill for diagnosing AI agents on Windows

- PowerShell-based checks for broken Python environments, PATH conflicts, MCP-provider misconfiguration; published as a reusable Claude Code skill.

TECHNICAL SKILLS

Languages:	Python, TypeScript, JavaScript, SQL, PowerShell, Bash
Backend:	FastAPI, Pydantic v2, async SQLAlchemy 2.0, Alembic, aiogram 3, Celery, REST APIs, JWT, RBAC
LLM / AI:	OpenAI SDK, Google Gemini, xAI Grok (OAuth + Agent Tools), MiMo, OpenRouter, RAG (sqlite-vec, fastembed), prompt engineering, structured output, function calling, multi-provider routers with circuit breaker
Frontend:	React 18, Next.js 14 (App Router), Tailwind CSS, TanStack Query, recharts
Data / Infra:	PostgreSQL 16, Redis, MinIO (S3), Docker, Docker Compose, Caddy, Hetzner, Linux
Quality:	pytest, testcontainers, respx, ruff, mypy (strict), pre-commit, gitleaks, structlog, Sentry
CI / Tools:	GitHub Actions, Git, Alembic, OpenAPI

EDUCATION

Universität Wien · BSc in Data Science	in progress
Statistics, machine learning, data processing, algorithms.	

LANGUAGES

Russian: Native · English: B2 · Deutsch: B2